

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice. (2 points each)

adaptation	genotype frequency	population
allometry	haploid	postzygotic barrier
allopatric	heterochrony	prezygotic barrier
allele frequency	heterozygote advantage	random mating
diploid	mutation	relative fitness
DNA code	natural selection	reproductive isolation
environment	non-random mating	stabilizing selection
gene flow	overall fitness	sympatric
genetic drift	paedomorphy	total offspring production
genotype	phenotype	

- 1) _____ Occurs when a population is not infinitely large.
 - 2) _____ Two alleles at a locus.
 - 3) _____ The combination of alleles for a gene in an organism.
 - 4) _____ New species evolve without geographic isolation.
 - 5) _____ Different reproductive behaviors prevent copulation.
 - 6) _____ Occurs when individuals move in *and* out of a population.
 - 7) _____ Creates new genetic variation within a population.
 - 8) _____ Can cause balancing selection in a population.
 - 9) _____ Different parts of the body develop at different rates at different times.
 - 10) _____ Trait that increases fitness in the current environment.
 - 11) _____ Inbreeding is an example of this cause of evolution.
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12) _____ $p + q$ represents this.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect. (2 points each)

- 13)** Which of these conditions should completely prevent the occurrence of natural selection in a population over time?
- A. The population lives in a habitat where there are no competing species present.
 - B. The environment is changing at a relatively slow rate.
 - C. All variation between individuals is due only to environmental factors.
 - D. The population size is large.
 - E. The population mates non-randomly.
- 14)** (Think carefully about this.) Organisms living in the northern rocky intertidal face competing selective forces. Larger individuals benefit by having a larger internal volume, which helps to conserve body temperature. However, smaller individuals are less likely to be dislodged by large waves from storms. Thus, selection may favor intermediate body size for any particular species. This is an example of
- A. stabilizing selection.
 - B. balancing selection.
 - C. natural selection.
 - D. directional selection.
 - E. disruptive selection.
- 15)** Krakatou is a volcanic island in Indonesia. An eruption in 1883 killed all living organisms on the island. Many of organismal populations that are present now tend to show low genetic variation. This is likely due to
- A. directional selection that favored traits that allow organisms to withstand volcanic activity.
 - B. a series of founder effects with different species recolonizing from surrounding islands.
 - C. The small population size causing random fluctuation of allele frequencies.
 - D. balancing selection that favored traits that allow organisms to withstand volcanic activity.
 - E. a genetic bottleneck caused by the eruption, followed by regrowth of the population.
- 16)** In a population with two alleles, *A* and *a*, that are in Hardy-Weinberg equilibrium, the frequency of allele *a* is 0.2. What is the frequency of heterozygotes?
- A. 0.02
 - B. 0.04
 - C. 0.16
 - D. 0.32
 - E. 0.4
- 17)** The biological species concept emphasizes
- A. gene flow between species.
 - B. the formation of cryptic species.
 - C. reproductive isolation between species.
 - D. the formation of morphologically distinct species.
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E. reproductive isolation between populations.

18) Two closely related species of maple trees occur together in the southeastern United States. Pollen from each species is distributed by the wind. If the pollen from one species lands on the stigma (female pollen receptor) of the other species but a protein interaction between the pollen and stigma prevents fertilization, then which reproductive barrier keeps the two maple species isolated?

- A. behavioral.
- B. gametic.
- C. mechanical.
- D. habitat.
- E. temporal.

19) Violation of these two Hardy-Weinberg assumptions always causes homozygosity to increase.

- A. Genetic drift and non-random mating.
- B. Genetic drift and natural selection.
- C. Gene flow and mutation.
- D. Mutation and natural selection.
- E. Non-random mating and gene flow.
- F. Natural selection and gene flow.

20) The rabbitsfoot mussel was once widespread throughout rivers of eastern North America but now remains in only a few very small populations. The decrease occurred rapidly due to habitat loss. Genetics studies show that the remaining populations have little or no genetic variation. The best explanation for this result is:

- A. Directional selection favored only one or two alleles.
- B. Balancing selection favored only one or two alleles.
- C. A bottleneck caused the loss of homozygosity.
- D. Genetic drift caused the loss of heterozygosity.
- E. Inbreeding caused an increase in heterozygosity.

21) According to Darwin's concept of descent with modification:

- A. All organisms overproduce offspring, leading to differential reproductive success.
- B. Species evolve from ancestral forms through the accumulation of different adaptations.
- C. All organisms overproduce offspring, leading to different adaptations.
- D. All species have evolved from a single common ancestor.
- E. Populations evolve by natural selection acting on individuals.
- F. Individuals with greater fitness will evolve more quickly than individuals with lower fitness.

22) Which mouse has the greatest relative fitness? (A clutch is the result of one breeding event.)

- A. One that lives for 3 years, has 3 clutches per year, and 7 offspring per clutch.
 - B. One that lives for 4 years, has 3 clutches per year, and 4 offspring per clutch.
 - C. One that lives for 3 years, has 2 clutches per year, and 6 offspring per clutch.
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- D. One that lives for 2 years, has 3 clutches per year, and 8 offspring per clutch.
 - E. One that lives for 3 years, has 3 clutches per year, and 5 offspring per clutch.
- 23)** Beetle pollinators of a particular plant are attracted to the bright orange flowers. The beetles not only pollinate the flowers, but they mate while inside of the flowers. A mutant version of the plant with red flowers becomes more common over time. A particular variant of the beetle prefers the red flowers to the orange flowers. Over time, these two beetle variants diverge from each other to such an extent that interbreeding is no longer possible. What kind of speciation has occurred in this example, and what has driven it?
- A. sympatric speciation; habitat isolation
 - B. allopatric speciation; behavioral isolation
 - C. sympatric speciation; gametic isolation
 - D. allopatric speciation; habitat isolation
 - E. sympatric speciation; temporal isolation
 - F. allopatric speciation; temporal isolation
- 24)** The overuse of drugs such as penicillin has caused an increase in the number of bacterial species that are resistant to the drugs. The most likely explanation for this result is:
- A. Most bacteria can resist drugs.
 - B. Bacteria began making proteins resistant to the drugs.
 - C. A few bacteria were naturally resistant and natural selection favored their increase.
 - D. The drugs caused mutations in the bacteria that gave them resistance.
 - E. Penicillin was not an effective drug.
- 25)** Two species of *Heliconius* butterflies occupy different habitats but the two species occur together in one area where their habitats are most similar. In this area, the two species often interbreed and hybrid offspring have been observed. Yet, biologists continue to classify the butterflies as two separate species. Which of the following observations would provide the best support for classifying the butterflies as two separate species?
- A. The two species are cryptic species.
 - B. The hybrids are cryptic species.
 - C. The hybrids can't reproduce.
 - D. The two species are morphological species, not biological species.
 - E. The two species are reproductively isolated because they occupy different habitats.
- 26)** Over time, the movement of people among all countries on Earth has steadily increased. This has altered the course of human evolution by increasing
- A. natural selection.
 - B. genetic drift.
 - C. gene flow.
 - D. non-random mating.
 - E. geographic isolation.
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- 27)** A species of sport fish reaches an average adult length of 100 cm. Fishing regulations require that only individuals longer than 75 cm may be kept. Shorter fish must be released because they have greater reproductive output relative to large individuals. Based on your knowledge of natural selection, you should predict that the average length of the adult fish population will
- A. remain unchanged.
 - B. gradually increase.
 - C. gradually decline.
 - D. rapidly decline.
 - E. rapidly increase.

Rock on!

Circle True or False. Read each statement completely before answering. (2 points each)

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|------------|------|-------|---|
| 28) | True | False | Gene flow is enough to stop the effects of genetic drift in a large population. |
| 29) | True | False | Natural selection by genetic drift is most common in small populations. |
| 30) | True | False | One assumption of the Hardy-Weinberg equations is that all individuals have an equally likely chance of migrating to another population. |
| 31) | True | False | Sexual dimorphism is most often the result of directional selection. |
| 32) | True | False | One difference between intrasexual and intersexual selection is that intrasexual selection does not cause sexual dimorphism. |
| 33) | True | False | If $p^2 = 0.16$, then total homozygosity in the population is 0.52. |
| 34) | True | False | Genetic variation among individuals in a population is necessary for natural selection to act upon individuals in that population. |
| 35) | True | False | Heterochrony can be defined as the retention of juvenile characteristics in the adult. |
| 36) | True | False | Microevolution and macroevolution are caused by the same underlying process: change in genetic variation in a population over time. |
| 37) | True | False | If a trait is energetically costly and provides no survival advantage, then genetic drift will cause that trait to disappear from the population over many generations. |
| 38) | True | False | The smallest unit of evolution is the individual. |
| 39) | True | False | Adaptations cause descent with modification. |
| 40) | True | False | Overproduction of offspring by every individual in a population is necessary for evolution to occur. |
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You may use a calculator to solve the following problem but you must **show your work** in the space provided. You may **round to 3 decimal places after each step or after the final calculation**. Any other rounding will result in a 3 point deduction. Failure to show your work will result in a 10 point deduction, even if your answers are correct.

- 41)** (15 points) Fruit flies use an enzyme, alcohol dehydrogenase (*adh*), to help metabolize rotting fruit. *adh* has two alleles, fast (F) and slow (S). A genetic sample of 1553 fruit flies found the following number of individual with each genotype is provided in the first table.

Use the information provided in the table at the right to fill in the two tables below with the following information: actual allele frequencies, observed genotype frequencies, and expected genotype frequencies. Do not calculate the expected number of individuals and do not perform a χ^2 test. The population is not in Hardy-Weinberg Equilibrium.

Genotype	<i>N</i>
FF	301
FS	924
SS	328

Allele Frequency		Observed Frequency	Expected Frequency
F	_____	FF	_____
S	_____	FS	_____
		SS	_____

- 42)** (5 points) The fruit fly population (from the previous page) is not in Hardy-Weinberg equilibrium because the sample contains an excess of heterozygotes and fewer homozygotes than expected. You suspect natural selection might explain the differences between observed and expected frequencies. Which mode of selection would explain the heterozygote excess? Explain why.
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